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Q1(c): Random 6x9 Matrix System

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Generating random 6x9 matrix A with random entries...

Generating random seed vector x_{seed} ...

Computing $b = A \times x_{\text{seed}}$ to ensure consistency...

Random 6x9 Matrix A:

[2.0, -3.0, -3.0, 2.0, -1.0, -2.0, -2.0, -2.0, 2.0]
[-3.0, 2.0, 2.0, 1.0, -3.0, 1.0, 0.0, -3.0, -3.0]
[-3.0, -2.0, -2.0, 1.0, 1.0, -3.0, 1.0, -2.0, 2.0]
[2.0, 2.0, 1.0, 0.0, -2.0, 0.0, 1.0, -1.0, 3.0]
[3.0, -3.0, 3.0, 3.0, -2.0, 2.0, 0.0, -1.0, -1.0]
[-2.0, -2.0, 3.0, -1.0, -3.0, -3.0, 0.0, -3.0, -1.0]

Vector b: [-2.0, -8.0, -16.0, 12.0, 11.0, -28.0]

REF [A|b]:

[2.0, -3.0, -3.0, 2.0, -1.0, -2.0, -2.0, -2.0, 2.0, -2.0]
[0.0, -2.5, -2.5, 4.0, -4.5, -2.0, -3.0, -6.0, 0.0, -11.0]
[0.0, 0.0, -1.0, 6.0, -10.0, -2.0, -3.0, -11.0, 1.0, -8.0]
[0.0, 0.0, 0.0, -6.4, 11.2, -0.8, 5.8, 10.6, 5.0, 9.6]
[0.0, 0.0, 0.0, 0.0, 4.0, -13.0, 18.0, -4.0, 32.0, 17.0]
[0.0, 0.0, 0.0, 0.0, 0.0, -29.312, 31.219, -14.656, 61.969, 6.687]

RREF [A|b]:

[1.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.19, 0.25, 0.981, 4.463]
[0.0, 1.0, 0.0, 0.0, 0.0, 0.0, 0.843, -0.125, 1.303, 4.355]
[0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.011, -0.0, 0.691, 1.382]
[0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 1.045, -0.625, 1.459, 4.668]
[0.0, 0.0, 0.0, 0.0, 1.0, 0.0, 1.039, 0.625, 1.129, 3.509]
[-0.0, -0.0, -0.0, -0.0, -0.0, 1.0, -1.065, 0.5, -2.114, -0.228]

Pivot columns (indices): [0, 1, 2, 3, 4, 5]

Non-pivot columns (free variables, indices): [6, 7, 8]

Number of pivots: 6

Number of free variables: 3

So columns without pivots become the free-variable columns.

This is why we get three parameters in the final answer.

Particular solution x_p (free vars = 0):

[4.4627, 4.355, 1.3817, 4.6684, 3.5085, -0.2281, 0.0, 0.0, 0.0]

Null space basis (solutions to $Ax = 0$):

n_1 : [-0.1903, -0.843, -1.0107, -1.0445, -1.0386, 1.065, 1.0, 0.0, 0.0]
 n_2 : [-0.25, 0.125, 0.0, 0.625, -0.625, -0.5, 0.0, 1.0, 0.0]
 n_3 : [-0.9813, -1.3025, -0.6908, -1.4592, -1.1293, 2.1141, 0.0, 0.0, 1.0]

General solution form:

$x = x_p + t_1 \cdot n_1 + t_2 \cdot n_2 + \dots + t_3 \cdot n_3$

where $t_1, t_2, \dots, t_3$ are free parameters

Verification:

Random coefficients t : [0.155, 0.409, -0.908]

Test solution x : [5.222, 5.459, 1.853, 6.088, 4.118, -2.188, 0.155, 0.409, -0.908]

Residual $\|Ax - b\|$: 0.0

Solution valid: YES

Since this residual is almost zero, the computed x satisfies $Ax = b$.

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prateeksrivastava-mac@prateeks-MacBook-Pro MFML_assignment_1 % python3 q1_linear_systems.py
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Q1(a) & Q1(b): HANDWRITTEN EXAMPLE (3x4 MATRIX)
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System of equations:

$$x_1 + 2x_2 + x_3 + x_4 = 5$$

$$2x_1 + 5x_2 + 3x_3 + 2x_4 = 12$$

$$x_1 + x_2 + 2x_3 + x_4 = 6$$

Original Matrix A:

[1, 2, 1, 1]

[2, 5, 3, 2]

[1, 1, 2, 1]

Vector b: [5, 12, 6]

Augmented Matrix [A|b]:

[1, 2, 1, 1, 5]

[2, 5, 3, 2, 12]

[1, 1, 2, 1, 6]

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Q1(a): REF AND RREF COMPUTATION
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REF [A|b]:

[1, 2, 1, 1, 5]

[0.0, 1.0, 1.0, 0.0, 2.0]

[0.0, 0.0, 2.0, 0.0, 3.0]

RREF [A|b]:

[1.0, 0.0, 0.0, 1.0, 2.5]

[0.0, 1.0, 0.0, 0.0, 0.5]

[0.0, 0.0, 1.0, 0.0, 1.5]

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Q1(b): PIVOT ANALYSIS AND SOLUTIONS
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Pivot columns (indices): [0, 1, 2]

Non-pivot columns (free variables, indices): [3]

Number of pivots: 3

Number of free variables: 1

Matrix rank: 3 (number of pivots)

Free variables: 1 (columns [3])

Particular solution x_p (free vars = 0):

[2.5, 0.5, 1.5, 0.0]

Null space basis (solutions to $Ax = 0$):

n_1: [-1.0, -0.0, -0.0, 1.0]

General solution form:

$$x = x_p + t_1 \cdot n_1$$

where $t_1, t_2, \dots$ are free parameters

Verification:

Test solution ($x_4 = 0$): [2.5, 0.5, 1.5, 0.0]

$A \cdot x = [5.0, 12.0, 6.0]$

Expected b = [5, 12, 6]

Residual $\|Ax - b\|$: 0.0

Solution is CORRECT!

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HANDWRITTEN EXAMPLE COMPLETED - READY FOR SCREENSHOTS
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